

Foundation (Jalapeno)

- Q1.** Solve the system: $x^2 + y^2 = 4$, $x - y = 0$. What is the value of x ?
(A) 0
(B) 1
(C) 2
(D) -1
(E) -2
- Q2.** A basketball player's score x and time y are related by $x = 2y^2 - 3y + 1$ and $y = x - 2$. What is the score when $y = 2$?
(A) 3
(B) 5
(C) 7
(D) 9
(E) 11
- Q3.** The equations $y = x^2 - 2x - 3$ and $y = 2x + 1$ represent two curves. What is the x -coordinate of their intersection?
(A) -1
(B) 1
(C) 2
(D) 3
(E) -2
- Q4.** If the system $x^2 - 4y = 0$ and $y^2 - x = 0$ has a solution (a, b) , what is $a + b$?
(A) 0
(B) 1
(C) 2
(D) 3
- (E) 4
- Q5.** For what value of x do the graphs of $y = x^2 + 2$ and $y = 2x$ intersect?
(A) 0
(B) 1
(C) 2
(D) -1
(E) -2
- Q6.** A football player runs a distance x in y minutes. If $y = 2x^2$ and $x = 2y - 4$, what is the distance when $y = 4$?
(A) 0
(B) 2
(C) 4
(D) 6
(E) 8
- Q7.** Solve for x : $x^2 + 4y^2 = 4$ and $x + y = 1$. What is x ?
(A) 0
(B) 1
(C) 2
(D) -1
(E) -2
- Q8.** If the system of equations $y = x^2 - 3$ and $y = x + 2$ has one solution, what is the value of x ?
(A) -1
(B) 1
(C) 2
(D) -2
(E) -5

Open Ended Questions (FRQ)

- Q9.** Graph the system $y = x^2$ and $y = 2x$. Label the points of intersection. What are their coordinates?
- Q10.** Solve the system $x^2 - 2y = 0$, $y^2 - x = 3$. Show all steps and express x and y as ordered pairs.

Chef's Tips

- Read each question carefully.
- Use the space provided for calculations.
- Show all steps in free response questions.